HOW TO BECOME A CRITICAL THINKER: THE ART OF READING SCIENTIFIC PAPERS

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Wake Forest University School of Medicine
Academic Core of ADVOCATEHEALTH

Overview

- Introduction
- Why do we read scientific papers ?
- Structure of a scientific paper
- Types of scientific papers
- How to decide which scientific papers to read?
- Reading effectively

Why do we read scientific papers?

Provide	general background information (rigor of prior research)
Provide	current information on a particular research field
Contain	detailed and useful methodology
Teach	you how to write
Identify	“gaps”

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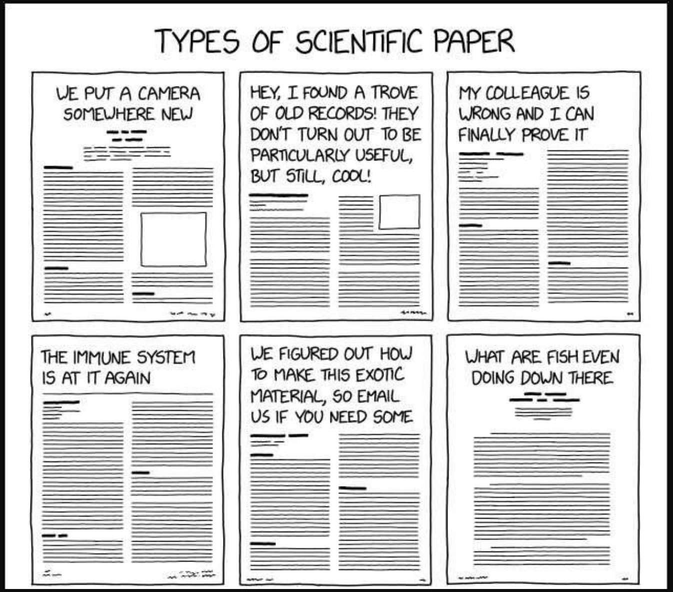
Types of Scientific Literature



Primary literature

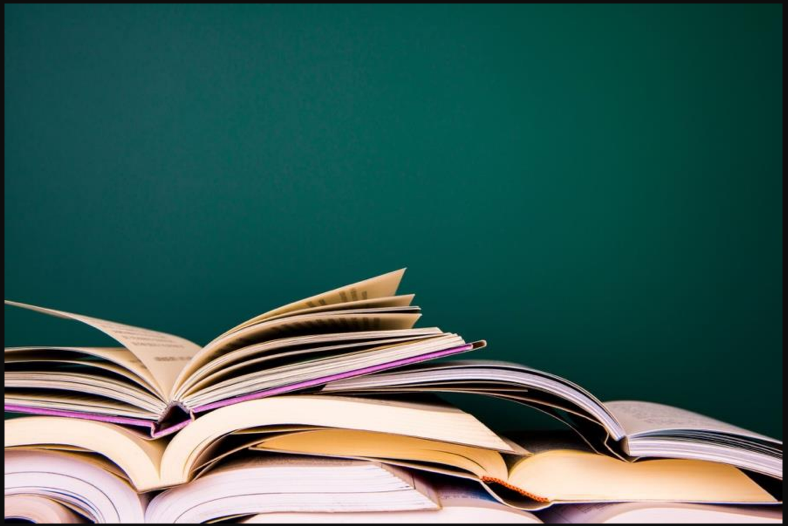
- Original articles
- Case reports
- Technical notes

Secondary literature

- Review articles
- Manuals

Tertiary literature

- Books, textbooks, encyclopedias



Types of Scientific Literature: **Preprints**

- arXiv (for physics, mathematics, computer science, and related fields)
- bioRxiv (for the life sciences)

  bioRxiv
  https://www.biorxiv.org
  bioRxiv.org - the preprint server for Biology
- medRxiv (for health sciences)

  medRxiv
  https://www.medrxiv.org
- ChemRxiv (for chemistry)
- SocArXiv (for social sciences)
- Preprints.org

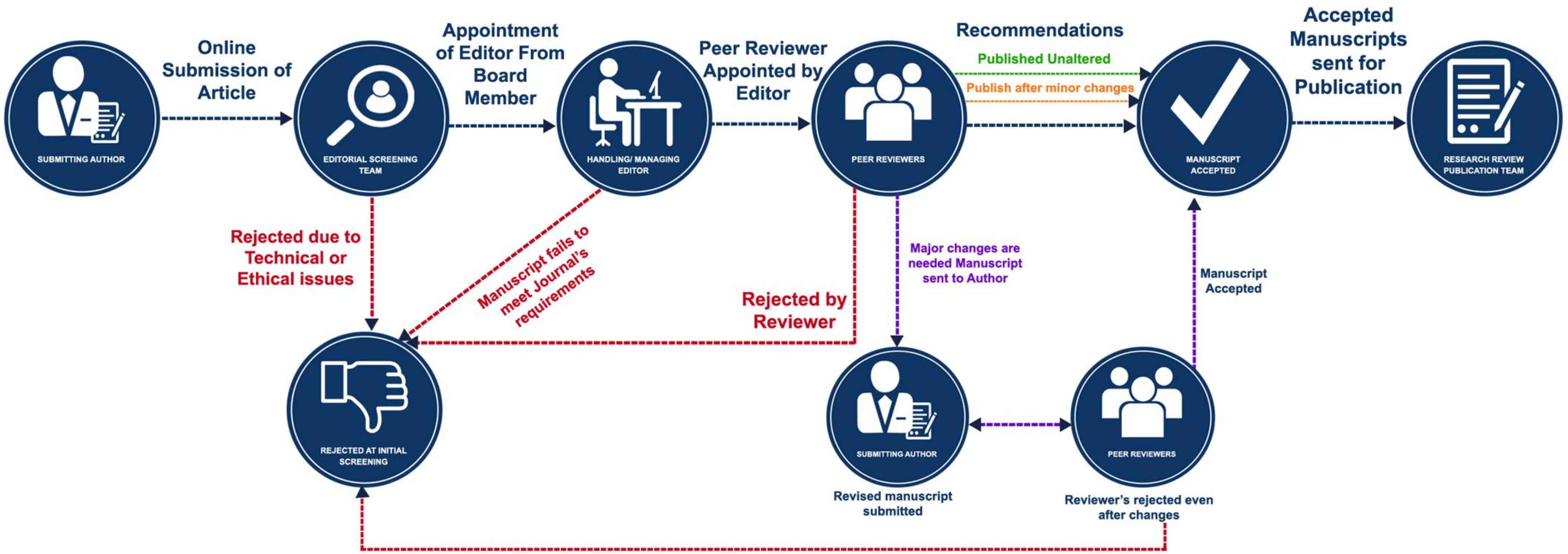
Good source of information:

- starting point for open discussion
- rapid dissemination of research
- wider public discussion
- help establish research priority

Caution:

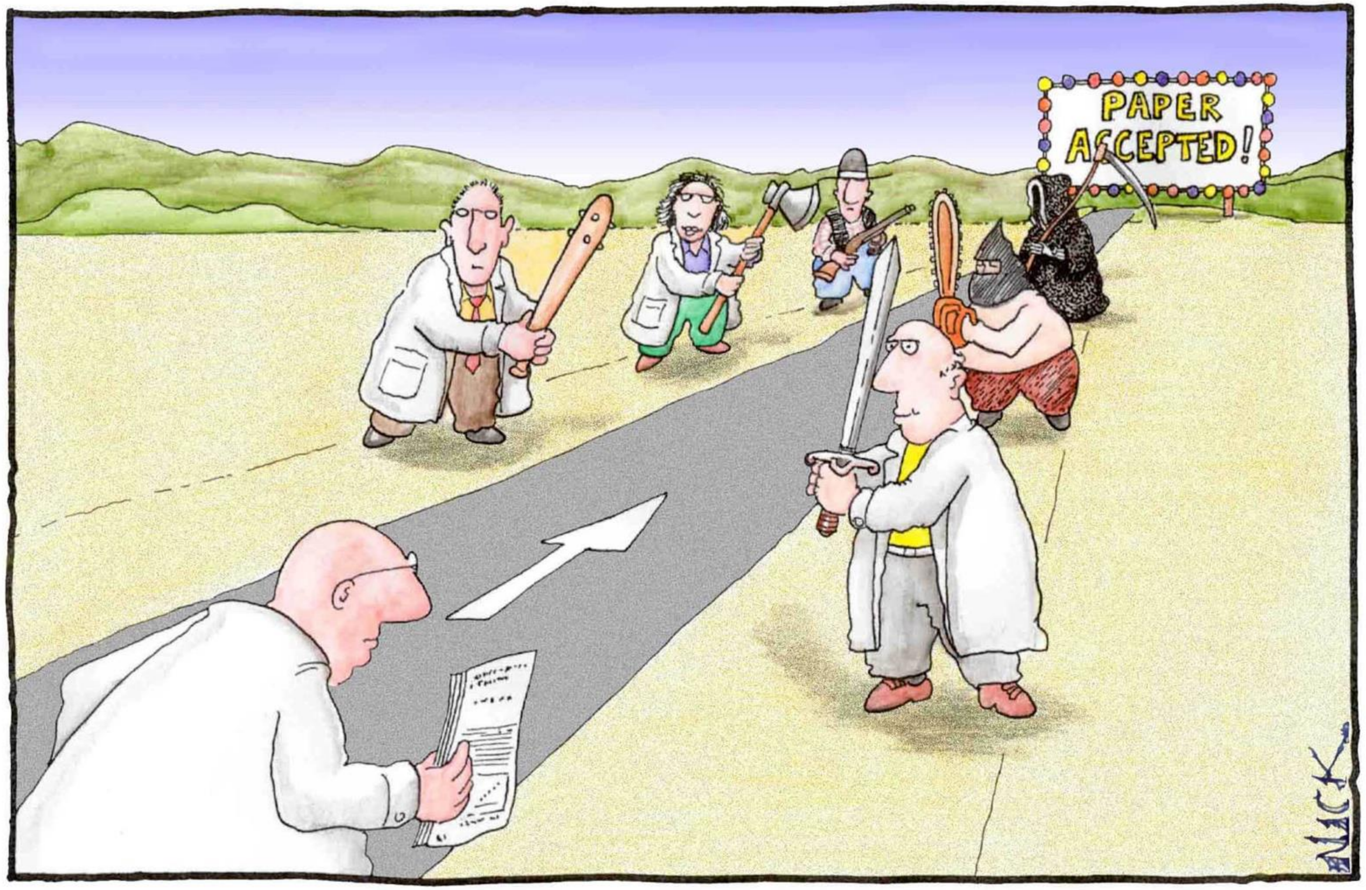
- **not peer reviewed**-no replacement for peer-reviewed publications
- Not reliable on for clinical practice or public health decision-making

Peer Review



SUBMITTING AUTHOR
Online Submission of Article
EDITORIAL SCREENING TEAM
Appointment of Editor From Board Member
HANDLING/ MANAGING EDITOR
Peer Reviewer Appointed by Editor
PEER REVIEWERS
Recommendations
Published Unaltered
Publish after minor changes
MANUSCRIPT ACCEPTED
Accepted Manuscripts sent for Publication
RESEARCH REVIEW PUBLICATION TEAM
Rejected due to Technical or Ethical issues
Manuscript fails to meet Journal's requirements
Rejected by Reviewer
Major changes are needed Manuscript sent to Author
Manuscript Accepted
REJECTED AT INITIAL SCREENING
SUBMITTING AUTHOR
Revised manuscript submitted
PEER REVIEWERS
Reviewer's rejected even after changes

Peer Review



Most scientists regarded the new streamlined peer-review process as 'quite an improvement.'

Featured image retrieved from: Science and Ink with credit to Nick Kim. See References

Structure of a Scientific Paper

Conventional structure

- Title
- Abstract
- Introduction
- Methods
- Results
- Discussion
- Acknowledgements and references
- Supplementary materials (some manuscripts)

Variations in structure depending on journal or type of paper

- "Methods last"Journals
  - Examples: *Nature*, *Science*, *Cell*, and *PNAS* .
  - *Nature:* results and discussion is one section
- Multi-experiment Papers
- Review Papers
- Papers on New Methods/Technologies
- Narrative or Unconventional Formats
- etc...

Introduction —

What is the background of your study?

Methods —

How did you conduct your study?

Results —

What did you find?

and

Discussion —

What do your results mean?

Title

- Keywords that describe the topic of your article
- Emphasis on most important aspect of your study
- Impact words – show the significance of your study

- Concise and descriptive

Breast Cancer Research and Treatment (2018) 172:69–82
https://doi.org/10.1007/s10549-018-4884-x

PRECLINICAL STUDY

CrossMark

Combination of anthracyclines and anti-CD47 therapy inhibit invasive breast cancer growth while preventing cardiac toxicity by regulation of autophagy

Yismeilin R. Feliz-Mosquea[1] · Ashley A. Christensen[1] · Adam S. Wilson[1] · Brian Westwood[1] · Jasmina Varagic[1,5] · Giselle C. Meléndez[3,4,5] · Anthony L. Schwartz[6] · Qing-Rong Chen[7] · Lesley Mathews Griner[8] · Rajarshi Guha[8] · Craig J. Thomas[8] · Marc Ferrer[8] · Maria J. Merino[6] · Katherine L. Cook[1,2,4,5] · David D. Roberts[6] · David R. Soto-Pantoja[1,2,4,5]

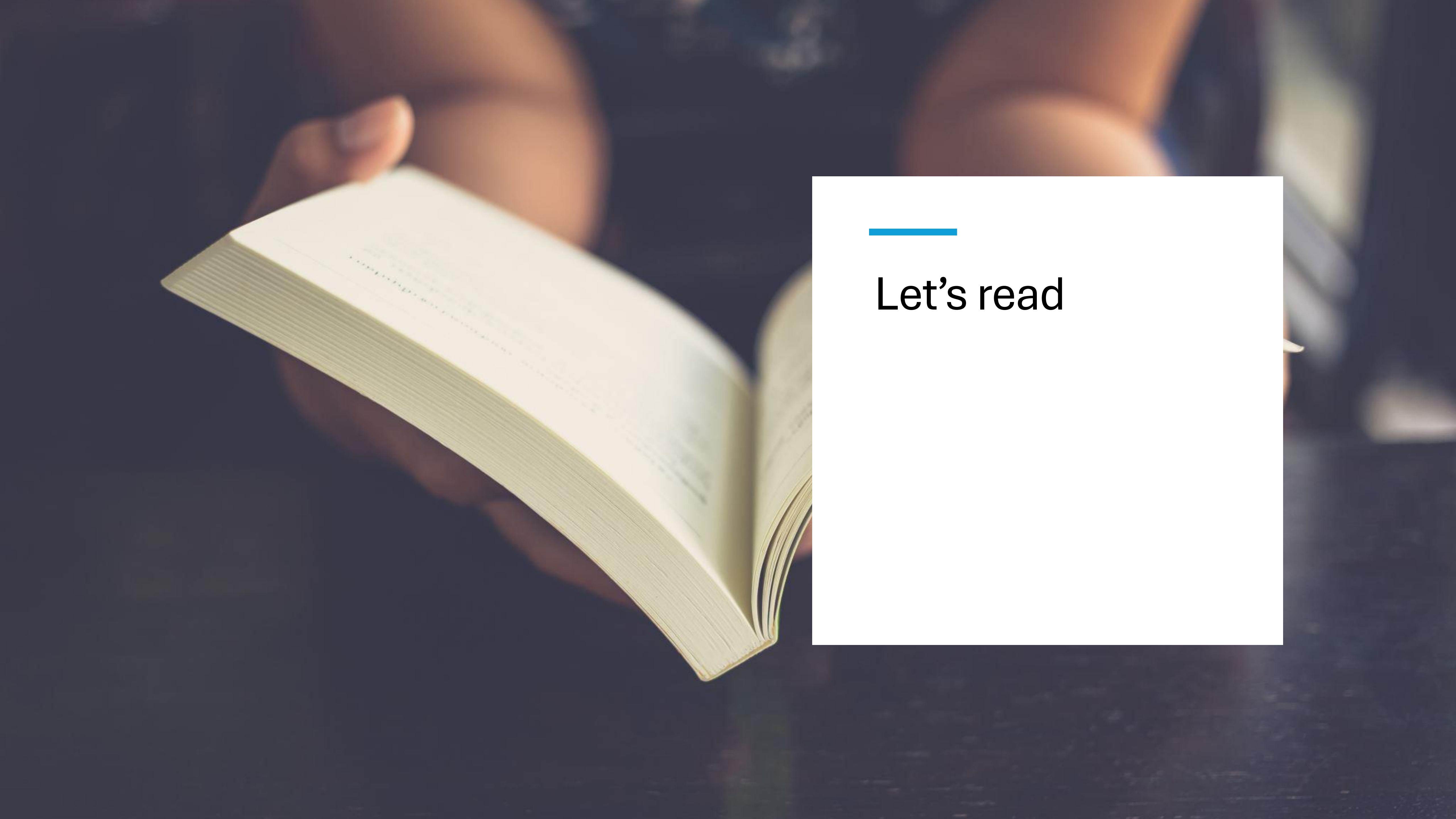
Does the authorship order matter? Does it mean anything?

Abstract

Brief summary: concise and compelling summary of your study

Parallels paper structure (IMRaD)

- Introduction/Background
- Methods
- Results / Major Findings
- Discussion/Conclusions and Significance



Let’s read

Abstract

Background A perennial challenge in systemic cytotoxic cancer therapy is to eradicate primary tumors and metastatic disease while sparing normal tissue from off-target effects of chemotherapy. Anthracyclines such as doxorubicin are effective chemotherapeutic agents for which dosing is limited by development of cardiotoxicity. Our published evidence shows that targeting CD47 enhances radiation-induced growth delay of tumors while remarkably protecting soft tissues. The protection of cell viability observed with CD47 is mediated autonomously by activation of protective autophagy. However, whether CD47 protects cancer cells from cytotoxic chemotherapy is unknown.

Methods We tested the effect of CD47 blockade on cancer cell survival using a 2-dimensional high-throughput cell proliferation assay in 4T1 breast cancer cell lines. To evaluate blockade of CD47 in combination with chemotherapy in vivo, we employed the 4T1 breast cancer model and examined tumor and cardiac tissue viability as well as autophagic flux.

Results Our high-throughput screen revealed that blockade of CD47 does not interfere with the cytotoxic activity of anthracyclines against 4T1 breast cancer cells. Targeting CD47 enhanced the effect of doxorubicin chemotherapy in vivo by reducing tumor growth and metastatic spread by activation of an anti-tumor innate immune response. Moreover, systemic suppression of CD47 protected cardiac tissue viability and function in mice treated with doxorubicin.

Conclusions Our experiments indicate that the protective effects observed with CD47 blockade are mediated through upregulation of autophagic flux. However, the absence of CD47 in did not elicit a protective effect in cancer cells, but it enhanced macrophage-mediated cancer cell cytolysis. Therefore, the differential responses observed with CD47 blockade are due to autonomous activation of protective autophagy in normal tissue and enhancement immune cytotoxicity against cancer cells.

Abstract

Background: Dietary recommendations may underestimate the protein older adults need for optimal bone health. This study sought to determine associations of protein intake with bone mineral density (BMD) and fracture among community-dwelling White and Black older adults.

Method: Protein as a percentage of total energy intake (TEI) was assessed with a Food Frequency Questionnaire in 2160 older adults (73.5 ± 2.8 years; 51.5% women; 35.8% Black) in the Health, Aging, and Body Composition prospective cohort. Hip, femoral neck, and whole body BMD was assessed by dual-energy x-ray absorptiometry at baseline and 4 years, and lumbar trabecular, cortical, and integral BMD was assessed by computed tomography at baseline and 5 years. Fragility fractures over 5 years were adjudicated from self-report data collected every 6 months. Associations with tertiles of protein intake were assessed using analysis of covariance for BMD and multivariate Cox regression for fracture, adjusting for confounders.

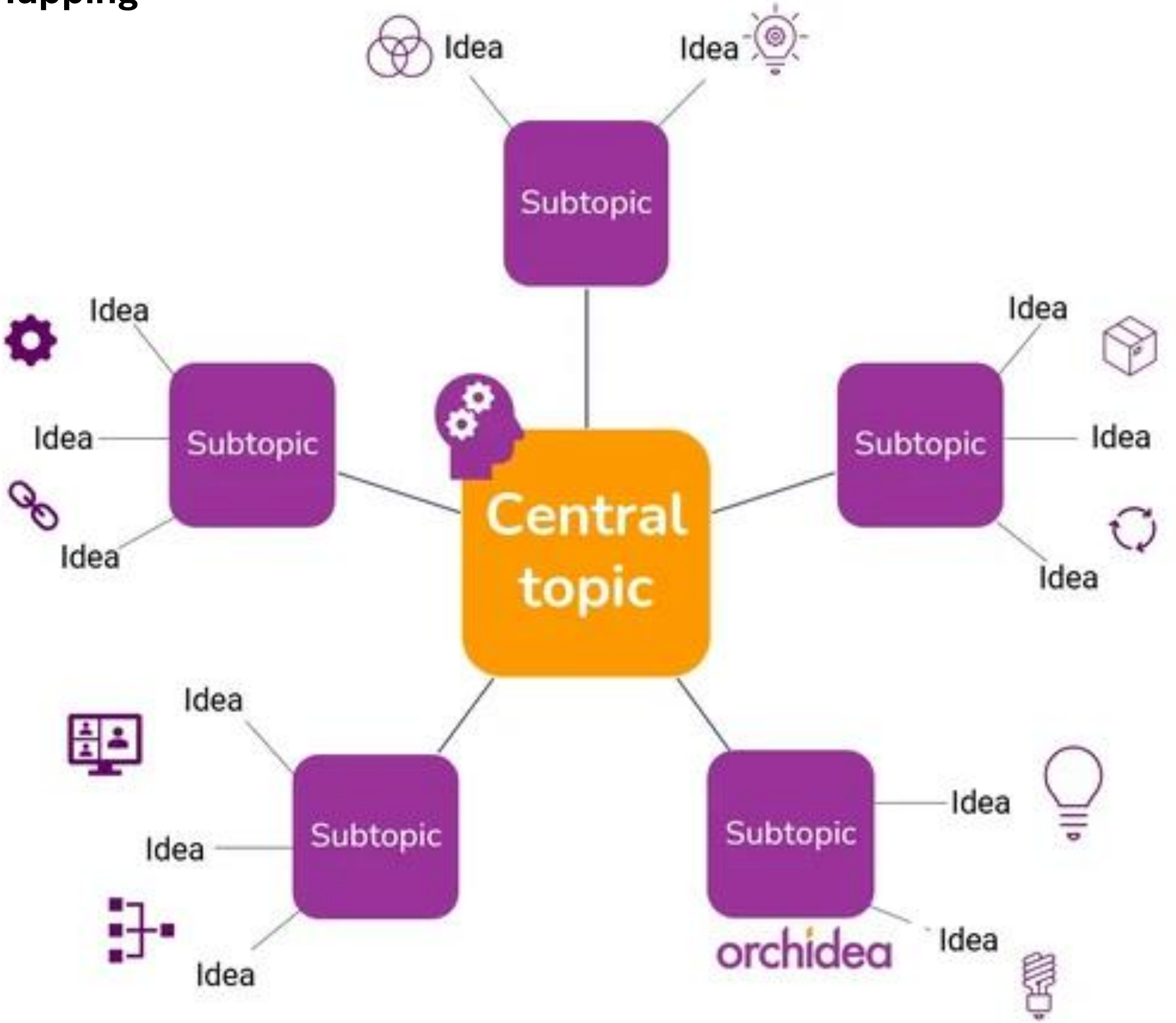
Results: Participants in the upper protein tertile (≥15% TEI) had 1.8%–6.0% higher mean hip and lumbar spine BMD compared to the lower protein tertile ($< .05$). Protein intake did not affect change in BMD at any site over the follow-up period. Participants in the upper protein tertile had a reduced risk of clinical vertebral fracture over 5 years of follow-up (hazard ratio: 0.36 [95% confidence interval: 0.14, 0.97] vs lower protein tertile, $p = .04$).

Conclusions: Older adults with higher protein intake (≥15% TEI) had higher BMD at the hip, whole body, and lumbar spine, and a lower risk of vertebral fracture. Keywords: Computed tomography (CT), Dual-energy x-ray absorptiometry (DXA), Food Frequency Questionnaire, Nutrition, Osteoporosis[2]

Abstract

This study evaluates the concordance between RNA sequencing (RNA-Seq) and NanoString technologies for gene expression analysis in non-human primates (NHPs) infected with Ebola virus (EBOV). A detailed comparison of both platforms revealed a strong correlation, with Spearman coefficients for 56 out of 62 samples ranging from 0.78 to 0.88. The mean and median coefficients were 0.83 and 0.85, respectively. Bland-Altman analysis confirmed high consistency across most measurements, with values falling within the 95% limits of agreement. Using a machine learning approach with the Supervised Magnitude-Altitude Scoring (SMAS) method trained on NanoString data, OAS1 was identified as a key gene signature for distinguishing RT-qPCR positive from negative samples. Remarkably, when used as the sole predictor in a logistic regression model, OAS1 maintained its predictive power on RNA-Seq data from the same cohort of EBOV-infected NHPs, achieving 100% accuracy in distinguishing infected from non-infected samples. OAS1 was also tested in a completely independent held-out test set, consisting of human monocyte-derived dendritic cells (DC) isolated and infected with different strains of the Ebola virus: wildtype (wt), VP35m, VP24m, along with a double mutant VP35m & VP24m, and again demonstrated a 100% accuracy rate in differentiating EBOV-infected from mock-infected samples, confirming its effectiveness as a predictive marker across diverse experimental setups and virus strains. Further differential expression analysis across both platforms identified 12 common genes (including ISG15, OAS1, IFI44, IFI27, IFIT2, IFIT3, IFI44L, MX1, MX2, OAS2, RSAD2, and OASL) that showed the highest levels of statistical significance and biological relevance. Gene Ontology (GO) analysis confirmed the involvement of these genes in key immune and viral infection pathways, highlighting their importance in EBOV infection. RNA-Seq uniquely identified genes such as CASP5, USP18, and DDX60, which are important in immune regulation and antiviral defense and were not detected by NanoString, demonstrating the broader detection capabilities of RNA-Seq. This study indicates a very strong agreement between RNA-Seq and NanoString platforms in gene expression analysis, with RNA-Seq displaying broader capabilities in identifying gene signatures.[1]

Mind Mapping



Introduction

- What topic is being investigated?
- What key background information is important to understand?

- Why is it being investigated?
- What are the knowledge gaps and the limitations in the rigor of the prior research?

- What is the hypothesis?
- What is your research question (and any aims)?

Materials and Methods/ Methods

- Describe materials and steps used to carry out experiments and analyze data
- Use very technical language
- Should be detailed enough for replication
  - In practice: descriptions highly compressed
  - To understand a method you may have to refer to earlier papers

Materials / Methods*

What and When

Study design and analysis design, if applicable; Time relative to when exposures and outcomes are assessed

Where

Setting

Who

(Inclusion and Exclusion Criteria): Study population relative to target population

How

Data sources, including how acquired

Definitions of major variables, especially exposures and outcomes

Why

Throughout, succinctly state why you made various assumptions and decisions about your methods. Do not hide or gloss over perceived weaknesses—own them

Analysis Plan

Descriptive statistics, among-group comparisons

State analytic purpose: descriptive, inferential, predictive

Modeling

- If multivariable (adjusted), provide details about directed acyclic graphs or other methods to identify model covariates
- Secondary/tertiary analyses, sensitivity analyses, etc

* Content mod, thank you to Dr. Andrew South

Results

Description of experiments and data

- References the data shown in figures and tables
- Often includes several different approaches

Minimal interpretation of data

Do not include:

- background or methods.
- discussion of presented data.
- avoid citing references in this section. Only include if they support methods that you developed in the study or similar methods that have been reported in the li

- **Were the results expected?**
- **Do the results/findings support the authors' hypothesis?**
- **Are the statistical methods appropriate?**

Results

- Summary of study population
- Primary results
- Secondary results
- Exploratory results/sensitivity analyses as applicable

Discussion

Interpret	Relate	Place	Suggest
Interpret the data • What do the data mean? • How do the data support the conclusion? • What are the strengths? • What are the limitations of the experiments? • Does the study support or contradict what is already known?	Relate findings to previous reports	Place findings in a broader context	Suggest future directions

Discussion

- Summary of primary findings (do not repeat the numeric results)
- Compare/contrast results with those from similar studies that have been published (at least three)
- Provide 2–3 alternative hypotheses to explain why we observed what we did; discuss mechanisms
- What are the immediate next steps and future directions?
- If confirmed and externally validated, what will your findings mean for human health at various levels (clinic/hospital, health care system, public health/policy, etc.)
- Honest and comprehensive limitations of our study and brief highlight of our strengths
- Conclusion to tie the manuscript all together

Other Sections

Abstract

Method

Results

Discussion

Funding

Conflict of Interest

Supplementary Material

Acknowledgments

References

Associated Data

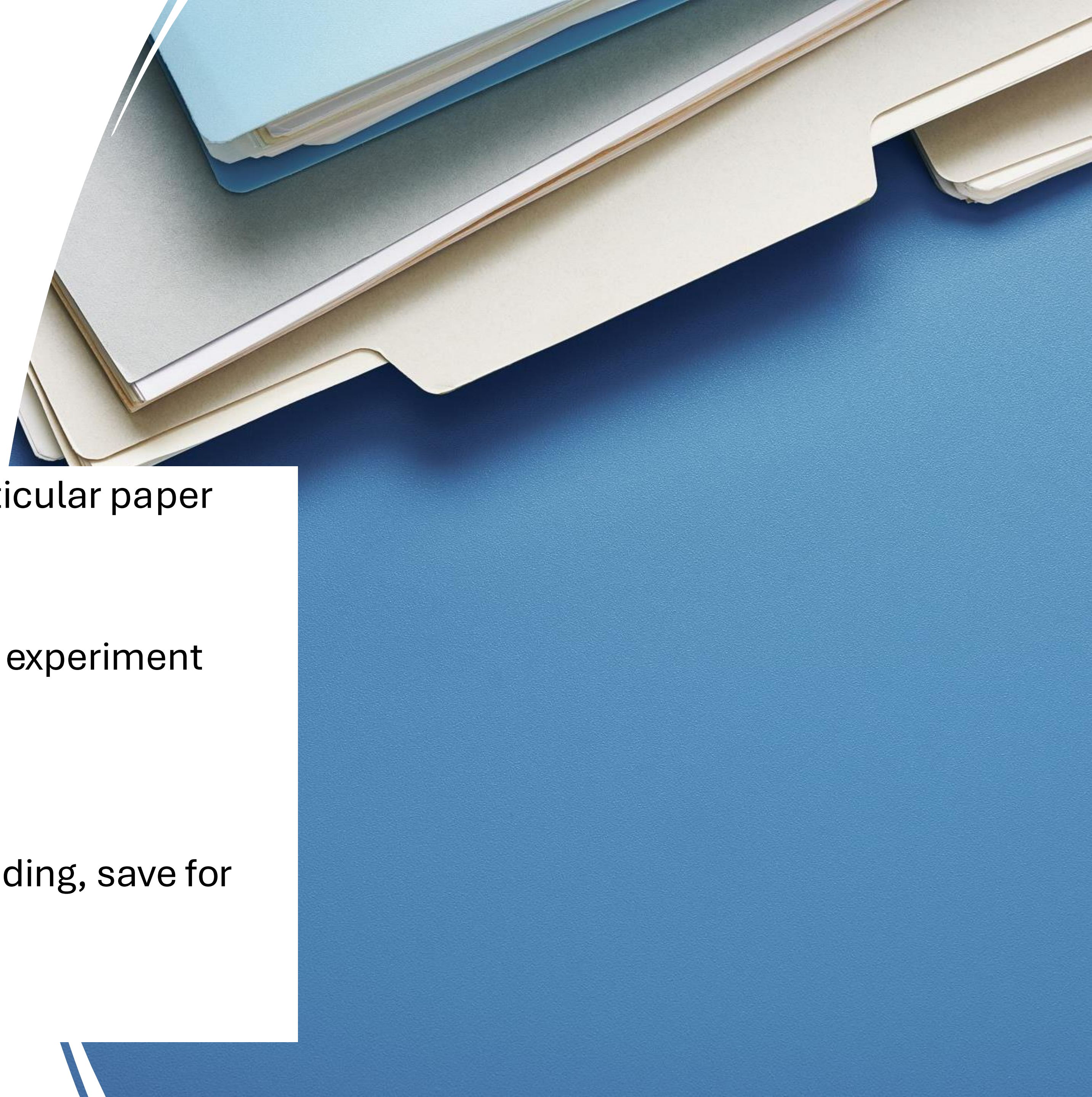
Deciding what papers to read

- Set a clear objective for reading a particular paper

  What do you want to get out of it?
    - Get and overview of a topic
    - Deeply understand a topic or experiment

- Identify the main finding of the paper
  - Read the title and abstract
  - Decide if you should continue reading, save for later, or skip it

- Impact factor



Read actively

Take notes as you read

Highlight major points

React to the main points discussed in the paper

Summarize what you read

Get an Overview

- Focus on the **title** and **abstract**
  - Remind yourself what you know about the topic
- **Skim the article** and
  - **analyze the document as a whole**
  - Section headings
  - Figures and tables
  - First and last paragraph of the introduction and discussion

Read in Depth

Find a reading order that works for you:

Introduction, Results, Discussion

Discussion, Introduction, Results

Figures, Discussion, Methods

Methods: can be read last or referred to as needed

Continuously refer to figures and tables

Look for keywords or phrases:

“surprising” “unexpected”

“in contrast with previous work”

“has seldom been addressed”

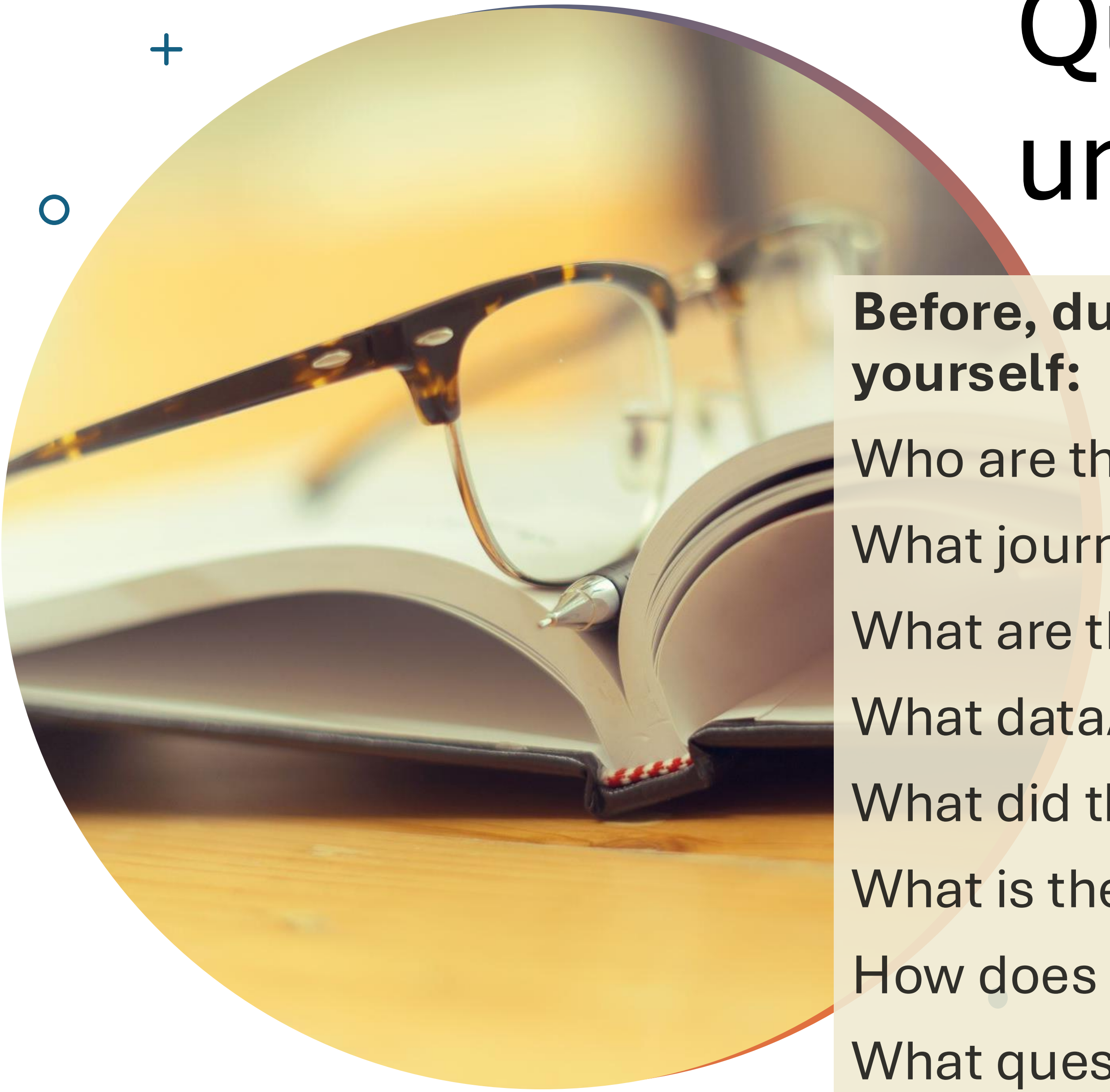
“we hypothesize”

“we propose”

“we introduce”

“the data suggest”

Question your understanding



Before, during, and after you start reading ask yourself:

Who are these authors?

What journal is this?

What are the main question(s) being asked?

What data/results emerged from the study?

What did the authors conclude?

What is the significance of these findings?

How does this article relate to others I have read?

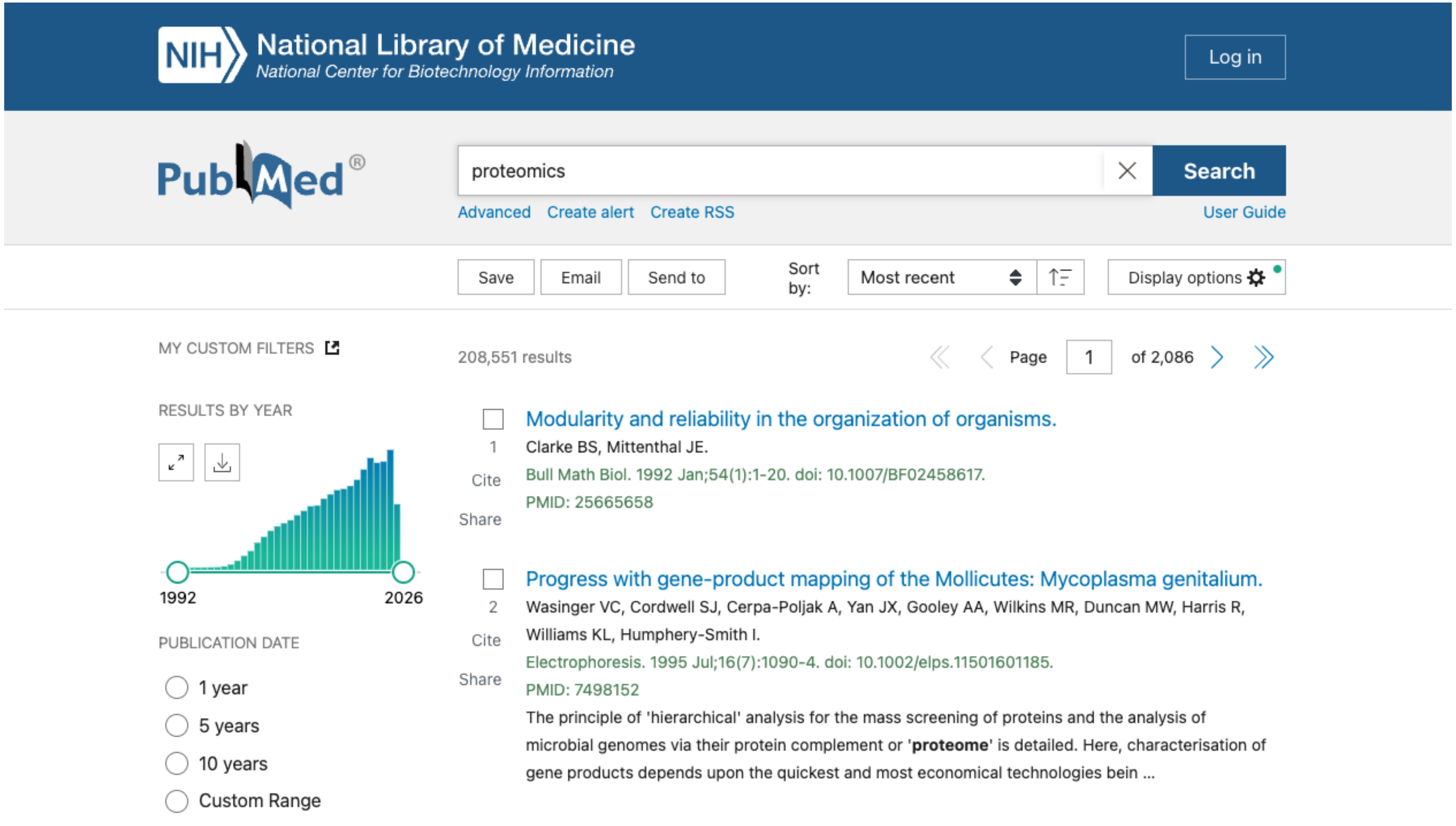
What questions are still unanswered?

Critically evaluate the paper

Examine the questions addressed in the paper: Descriptive, Comparative, or Analytical

- Examine:
  - Evidence and statistics
  - Conclusions
- Determine if the data support conclusions
  - Logical connection between data and conclusion?
- Any other interpretations?
- Relate the findings to what you already know
- Consider merits and limitations of the paper

(set to 100 results per page)



PubMed®

proteomics and obesity

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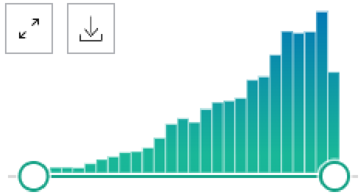
RESULTS BY YEAR



PUBLICATION DATE

- 1 year
- 5 years
- 10 years
- Custom Range

TEXT AVAILABILITY

- Abstract

2,836 results

Page 1 of 29

1 Cite Share

A proteome analysis of livers from obese (ob/ob) mice treated with the peroxisome proliferator WY14,643.

Edvardsson U, Alexandersson M, Brockenhuus von Löwenhielm H, Nyström AC, Ljung B, Nilsson F, Dahllöf B.

Electrophoresis. 1999 Apr-May;20(4-5):935-42. doi: 10.1002/(SICI)1522-2683(19990101)20:4/5<935::AID-ELPS935>3.0.CO;2-6.

PMID: 10344269

However, we wanted to map the effects of a therapeutic dose of a PPARalpha agonist in a disease model of insulin resistance and diabetes, the **obese** diabetic ob/ob mouse, by **proteomics**. Therefore, ob/ob mice, which have highly elevated levels of plasma triglycerides, ...

2 Cite Share

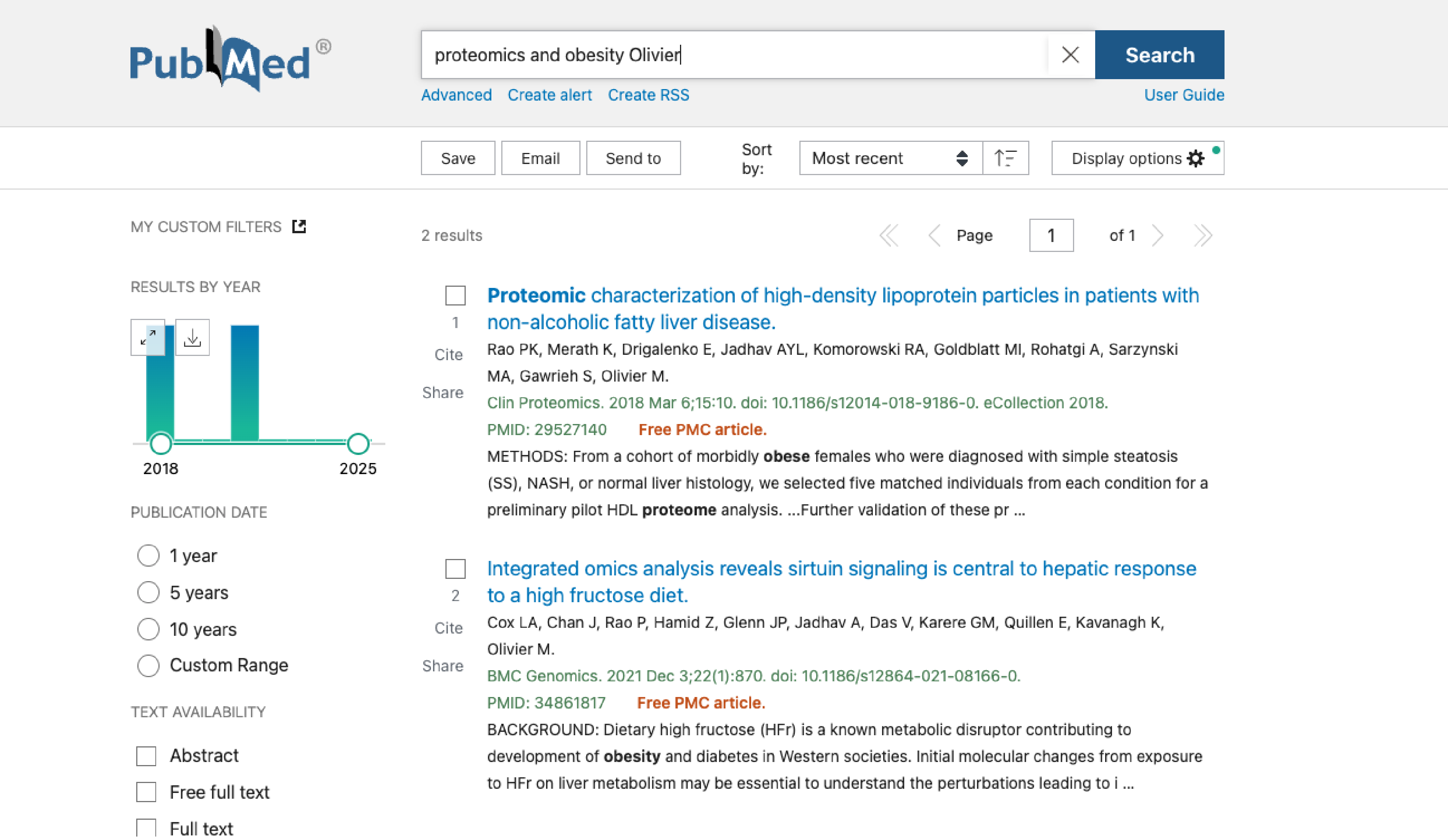
Research in the exercise sciences: where do we go from here?

Baldwin KM.

J Appl Physiol (1985). 2000 Jan;88(1):332-6. doi: 10.1152/jappl.2000.88.1.332.

PMID: 10642398 **Free article.** Review.

These successes will drive the emerging fields of functional genomics (the dissecting of a gene's identity and function) and **proteomics** (the study of the properties of proteins). Funding levels at the National Institutes of Health will likely increase in order to expand th ...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proteomics and obesity Olivier
Search
Advanced
Create alert
Create RSS
User Guide
Save
Email
Send to
Sort by:
Most recent
Display options
MY CUSTOM FILTERS
2 results
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RESULTS BY YEAR
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2025
PUBLICATION DATE
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5 years
10 years
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Abstract
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1
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Proteomic characterization of high-density lipoprotein particles in patients with non-alcoholic fatty liver disease.
Rao PK, Merath K, Drigalenko E, Jadhav AYL, Komorowski RA, Goldblatt MI, Rohatgi A, Sarzynski MA, Gawrieh S, Olivier M.
Clin Proteomics. 2018 Mar 6;15:10. doi: 10.1186/s12014-018-9186-0. eCollection 2018.
PMID: 29527140 Free PMC article.
METHODS: From a cohort of morbidly obese females who were diagnosed with simple steatosis (SS), NASH, or normal liver histology, we selected five matched individuals from each condition for a preliminary pilot HDL proteome analysis. ...Further validation of these pr ...
2
Cite
Share
Integrated omics analysis reveals sirtuin signaling is central to hepatic response to a high fructose diet.
Cox LA, Chan J, Rao P, Hamid Z, Glenn JP, Jadhav A, Das V, Karere GM, Quillen E, Kavanagh K, Olivier M.
BMC Genomics. 2021 Dec 3;22(1):870. doi: 10.1186/s12864-021-08166-0.
PMID: 34861817 Free PMC article.
BACKGROUND: Dietary high fructose (HFr) is a known metabolic disruptor contributing to development of obesity and diabetes in Western societies. Initial molecular changes from exposure to HFr on liver metabolism may be essential to understand the perturbations leading to i ...

Filters – refine search

- 1 year
- 5 years
- 10 years
- Custom Range

TEXT AVAILABILITY

- Abstract
- Free full text
- Full text

ARTICLE ATTRIBUTE

- Associated data

ARTICLE TYPE

- Books and Documents
- Clinical Trial
- Meta-Analysis
- Randomized Controlled Trial
- Review
- Systematic Review

See all article type filters

Additional filters –

ARTICLE LANGUAGE

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- Spanish

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SPECIES

- Humans
- Other Animals

SEX

- Female
- Male

AGE

- Child: birth-18 years
- Adult: 19+ years
- Aged: 65+ years

See all age filters

OTHER

- Exclude preprints
- MEDLINE

References

- The Different Types of Scientific Literature: http://www.um.edu.mt/__data/assets/file/0006/42981/The_different_types_of_scientific_literature.pdf
- Basic Structure and Types of Scientific Papers. Effective Medical Writing, Singapore Med J, 2008
- How to Read a Scientific Paper – University of Arizona: http://www.biochem.arizona.edu/classes/bioc568/papers.htm
- How to read a Scientific Article – Rice University: http://www.owlnet.rice.edu/~cainproj/courses/HowToReadSciArticle.pdf
- Efficient Reading of Papers in Science and Technology: http://www.cs.columbia.edu/~hgs/netbib/efficientReading.pdf
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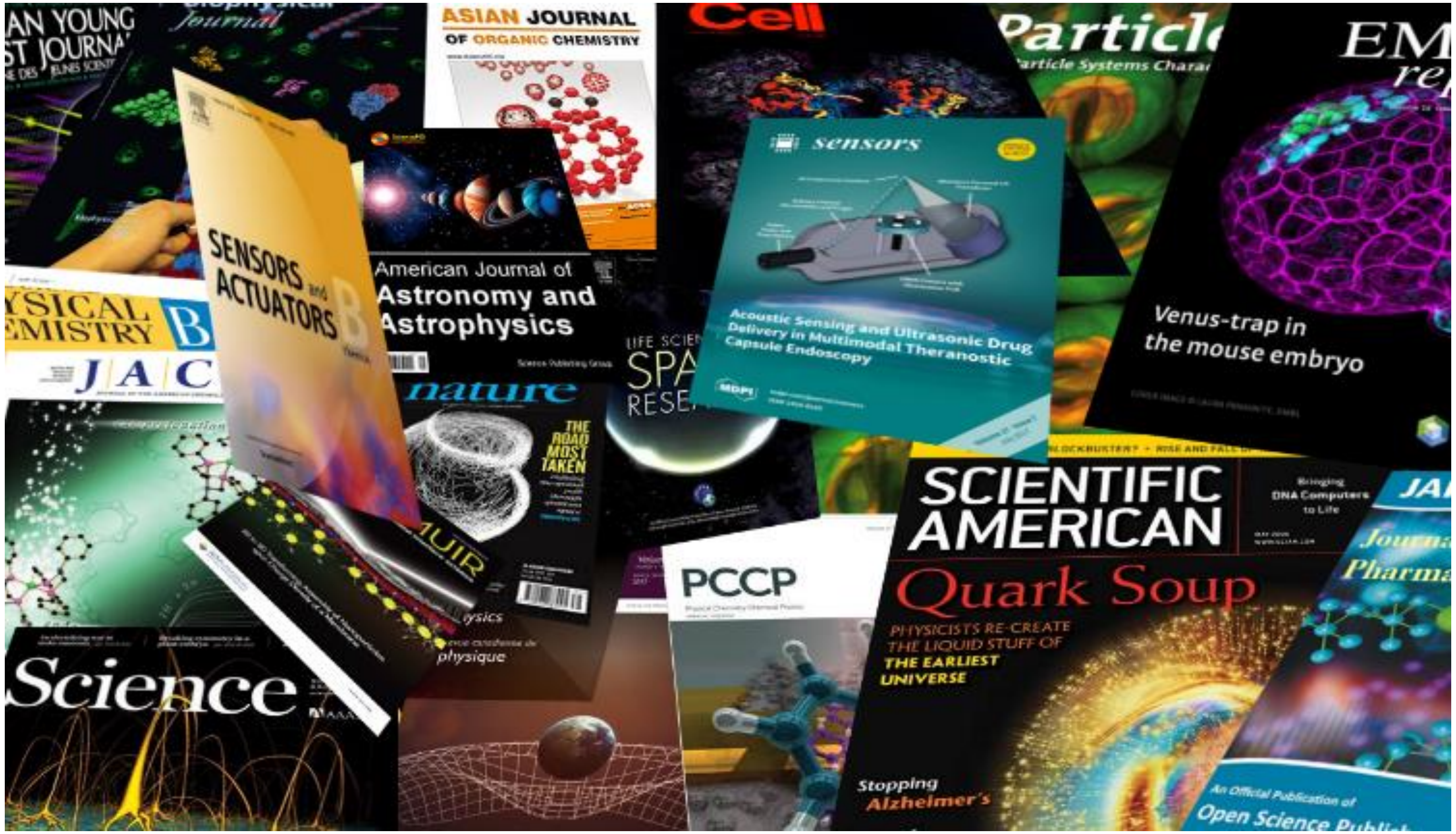
- [2]Weaver AA, Tooze JA, Cauley JA, Bauer DC, Tylavsky FA, Kritchevsky SB, Houston DK. Effect of Dietary Protein Intake on Bone Mineral Density and Fracture Incidence in Older Adults in the Health, Aging, and Body Composition Study. J Gerontol A Biol Sci Med Sci. 2021 Nov 15;76(12):2213-2222. doi: 10.1093/gerona/glab068. PMID: 33677533; PMCID: PMC8599066.
- [1]Rezapour M, Narayanan A, Mowery WH, Gurcan MN. Assessing concordance between RNA-Seq and NanoString technologies in Ebola-infected nonhuman primates using machine learning. BMC Genomics. 2025 Apr 10;26(1):358. doi: 10.1186/s12864-025-11553-6. PMID: 40211167; PMCID: PMC11983957.

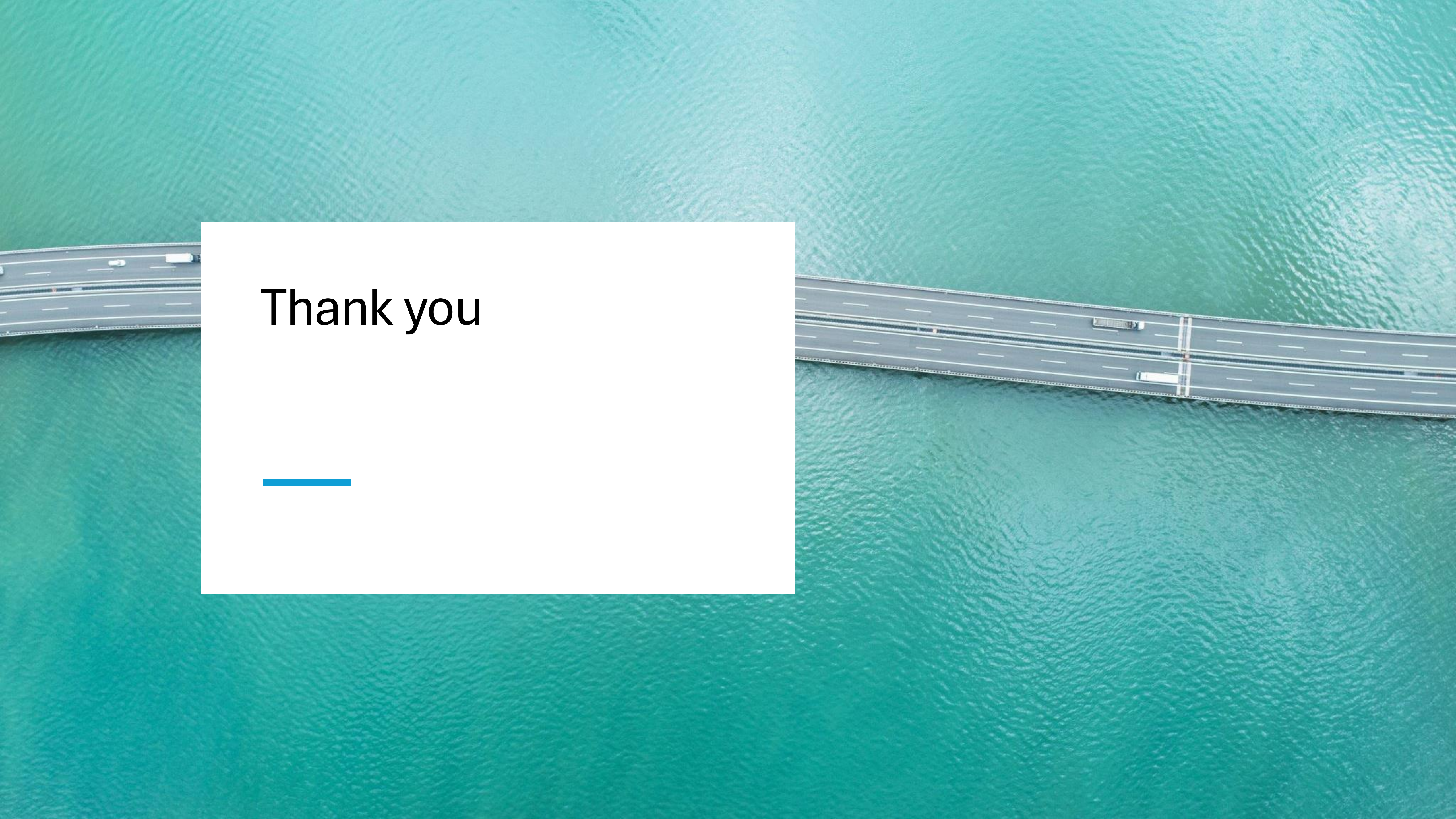
Preparing for your journal club…

(1st: June 18, 1 pm)



Journal Clubs (Cardiovascular and Aging)

- Students will be working in groups of 3.
- Each group of students will be assigned a journal article from program faculty.
- Each student will be assigned to present the introduction, methods or results (powerpoint).
- All 3 students will collectively present the discussion and conclusion.



Thank you